All Labeled Mutants vs. Rank-Based Comparator | Page 1/3

Comparator rule: if mutant rank = 1, show rank 2; otherwise show rank 1.

Observed quadrant counts and MUT/Q4 enrichment

Mutant rank	Plate	Known mutant (well | N | Q1/Q2/Q4)	MUT/Q4	Comparison rank	Comparison well (well | N | Q1/Q2/Q4)	MUT/Q4
1	200122_TRT13_TRT032_F1 R1	A08 | N=11,457 | 25/18/7,826	0.005495	2	C07 | N=10,688 | 26/6/8,968	0.003568
1	200130_TRT13_TRT054_(BR	C11 | N=14,546 | 87/33/10,528	0.0114	2	G02 | N=13,872 | 7/4/10,665	0.001031
1	200226_TRT23_TRT081	G11 | N=15,409 | 46/7/14,223	0.003726	2	H10 | N=15,028 | 48/6/12,635	0.004274
1	200306_TRT23_TRT095	C08 | N=15,764 | 52/1/15,605	0.003396	2	D02 | N=17,956 | 5/5/10,626	0.0009411
1	200309_TRT17_TRT003_BR	D10 | N=15,940 | 54/0/15,821	0.003413	2	G04 | N=14,302 | 34/0/14,182	0.002397
1	200319_TRT2_TRT073	E06 | N=16,780 | 13/17/5,089	0.005895	2	H07 | N=18,711 | 6/14/3,103	0.006445
1	201001_TRT20_TRT118	C05 | N=13,420 | 3/7/3,249	0.003078	2	B07 | N=12,930 | 39/0/12,427	0.003138
1	201015_TRT25_TRT138	D06 | N=10,589 | 97/5/9,495	0.01074	2	H12 | N=14,717 | 42/7/13,113	0.003737
1	201027_TRT25_TRT127	D06 | N=14,994 | 172/12/13,872	0.01326	2	H12 | N=14,241 | 10/1/13,488	0.0008155
1	210106_ARM3-TRT158	D09 | N=8,899 | 9/14/3,813	0.006032	2	B04 | N=9,854 | 1/8/5,192	0.001733
1	210121_TRT13_TRT008_(F1-R1)	G06 | N=13,588 | 26/9/12,354	0.002833	2	C10 | N=15,177 | 6/4/14,761	0.0006775
1	210218_TRT25_TRT26_TA175	D05 | N=13,564 | 59/5/12,578	0.005088	2	G10 | N=13,743 | 51/1/13,600	0.003824
1	210317_RF15_TA189	F12 | N=18,290 | 4/7/6,451	0.001705	2	G08 | N=18,629 | 2/6/6,993	0.001144
1	210531_TRT19_TA199	H02 | N=14,070 | 0/10/2,474	0.004042	2	B07 | N=9,789 | 0/0/0	NA
1	210601_TRT19_TA205_2.half	H04 | N=17,373 | 1/4/2,569	0.001946	2	F09 | N=17,612 | 6/0/6,711	0.0008941
1	210721_TRT61_TA209	A01 | N=18,638 | 90/4/17,582	0.005346	2	G02 | N=19,450 | 25/0/19,331	0.001293
1	210803_MIP054_TRT53_TA217	G09 | N=14,700 | 19/1/12,350	0.001619	2	H02 | N=15,504 | 10/1/12,885	0.0008537
1	210819_MIP-056_TAS1_TRT219	A01 | N=17,749 | 114/3/17,347	0.006745	2	G01 | N=16,941 | 137/0/16,612	0.008247
1	210902_TRT60_TA164+1	D12 | N=16,685 | 5/6/7,729	0.001423	2	D05 | N=11,253 | 3/2/4,482	0.001116
1	210913_TAS51_TA222_	B03 | N=15,829 | 10/20/6,647	0.004513	2	B05 | N=13,561 | 1/3/5,285	0.0007569
1	211001_TRT65_TA223	D08 | N=15,885 | 17/11/6,594	0.004246	2	D09 | N=16,206 | 7/7/6,419	0.002181
1	211007_TRT42_TA237	E04 | N=14,024 | 1/8/2,257	0.003988	2	H06 | N=15,579 | 0/0/0	NA

Independent MUT/WT occupancy reliability

Mutant rank	Plate	Known mutant: E(Q1)/E(Q2); Fisher p; warning	Comparison rank	Comparator: E(Q1)/E(Q2); Fisher p; warning
1	200122_TRT13_TRT032_F1 R1	29.5/13.5; p=0.14; OK	2	26.9/5.1; p=0.63; OK
1	200130_TRT13_TRT054_(BR	87.6/32.4; p=0.92; OK	2	8.5/2.5; p=0.29; OK
1	200226_TRT23_TRT081	49.1/3.9; p=0.11; OK	2	45.6/8.4; p=0.45; OK
1	200306_TRT23_TRT095	52.6/0.4; p=0.3; OK	2	5.9/4.1; p=0.54; OK
1	200309_TRT17_TRT003_BR	53.8/0.2; p=1; OK	2	33.8/0.2; p=1; OK
1	200319_TRT2_TRT073	9.1/20.9; p=0.16; OK	2	3.3/16.7; p=0.13; OK
1	201001_TRT20_TRT118	2.4/7.6; p=0.71; OK	2	37.6/1.4; p=0.4; OK
1	201015_TRT25_TRT138	92.4/9.6; p=0.13; OK	2	43.8/5.2; p=0.36; OK
1	201027_TRT25_TRT127	172.3/11.7; p=0.88; OK	2	10.4/0.6; p=0.45; OK
1	210106_ARM3-TRT158	9.9/13.1; p=0.83; OK	2	4.7/4.3; p=0.016; OK
1	210121_TRT13_TRT008_(F1-R1)	31.9/3.1; p=0.0028; OK	2	9.7/0.3; p=9.7e-05; WARN
1	210218_TRT25_TRT26_TA175	59.6/4.4; p=0.62; OK	2	51.7/0.3; p=0.3; OK
1	210317_RF15_TA189	3.9/7.1; p=1; OK	2	3.0/5.0; p=0.72; OK
1	210531_TRT19_TA199	1.8/8.2; p=0.23; OK	2	0.0/0.0; p=1; OK
1	210601_TRT19_TA205_2.half	0.7/4.3; p=0.55; OK	2	2.3/3.7; p=0.0031; OK
1	210721_TRT61_TA209	89.1/4.9; p=1; OK	2	24.9/0.1; p=1; OK
1	210803_MIP054_TRT53_TA217	16.8/3.2; p=0.35; OK	2	9.1/1.9; p=0.7; OK
1	210819_MIP-056_TAS1_TRT219	115.1/1.9; p=0.44; OK	2	135.4/1.6; p=0.41; OK
1	210902_TRT60_TA164+1	5.1/5.9; p=1; OK	2	2.0/3.0; p=0.39; OK
1	210913_TAS51_TA222_	12.6/17.4; p=0.36; OK	2	1.6/2.4; p=1; OK
1	211001_TRT65_TA223	11.7/16.3; p=0.053; OK	2	5.6/8.4; p=0.43; OK
1	211007_TRT42_TA237	1.4/7.6; p=1; OK	2	0.0/0.0; p=1; OK

N = called droplets; counts are Q1/Q2/Q4. E(Q1/Q2) uses marginal MUT/WT rates. Fisher p < 0.001 is a reliability warning, not an artifact or mutant-call label.

All Labeled Mutants vs. Rank-Based Comparator | Page 2/3

Comparator rule: if mutant rank = 1, show rank 2; otherwise show rank 1.

Observed quadrant counts and MUT/Q4 enrichment

Mutant rank	Plate	Known mutant (well | N | Q1/Q2/Q4)	MUT/Q4	Comparison rank	Comparison well (well | N | Q1/Q2/Q4)	MUT/Q4
1	211116_TRT63_TA250	H11 | N=13,958 | 34/0/13,642	0.002492	2	B03 | N=12,627 | 0/0/0	NA
1	220104_MIP-064_TRT63_TA253	D03 | N=14,748 | 115/8/13,878	0.008863	2	A08 | N=17,467 | 47/9/14,995	0.003735
1	220113_TRT76_TA241	A09 | N=17,997 | 15/3/12,882	0.001397	2	A03 | N=18,143 | 9/0/7,577	0.001188
1	220119_MIP-068-PS_TRT44_TA280	F01 | N=15,858 | 1/10/5,753	0.001912	2	C02 | N=14,295 | 0/1/4,747	0.0002107
1	220121_MIP-062-MC_TRT76_TA246	H02 | N=11,175 | 69/4/10,562	0.006912	2	G07 | N=13,430 | 12/0/12,780	0.000939
1	220125_TRT20_TA287	C10 | N=9,922 | 28/8/7,799	0.004616	2	A03 | N=16,347 | 0/3/2,694	0.001114
1	220215_MIP069_TRT19_TA284	A08 | N=17,430 | 2/9/2,352	0.004677	2	H08 | N=16,844 | 4/1/1,979	0.002527
1	220303_MIP-076-PS_TRT43_TA304	B03 | N=12,511 | 7/9/5,015	0.00319	2	C03 | N=14,341 | 6/1/11,671	0.0005998
1	220309_MIP059_TRT42_TA231s_	C05 | N=13,638 | 4/6/2,158	0.004634	2	E08 | N=11,489 | 1/3/4,449	0.0008991
1	220324_TRT50_TA299	G09 | N=17,696 | 21/10/13,487	0.002299	2	F01 | N=16,766 | 17/1/15,643	0.001151
1	220331_MIP-072_CQ_TRT23_TA293	B07 | N=14,668 | 9/7/6,087	0.002629	2	C08 | N=14,424 | 1/5/8,966	0.0006692
1	220408_MIP-067_FF28_TA274	D11 | N=16,054 | 5/37/6,387	0.006576	2	H11 | N=17,329 | 1/4/7,735	0.0006464
1	220427-MIP073CQ_TRT93_TA296_dd	G12 | N=14,087 | 1/8/3,757	0.002396	2	C03 | N=15,384 | 2/3/3,676	0.00136
1	220519_MIP_22_088_TRT50__20220518_144839_988	F02 | N=15,663 | 16/17/12,795	0.002579	2	A10 | N=15,996 | 10/2/12,955	0.0009263
1	220520_TRT19_TA330_	G04 | N=17,196 | 3/4/3,909	0.001791	2	B07 | N=14,916 | 0/0/0	NA
1	220531_MIP-082-MC_TRT76_TA341	C08 | N=17,280 | 64/6/14,984	0.004672	2	C01 | N=18,066 | 16/1/17,218	0.0009873
1	220531_MIP-085_TRT23_TA333_	A04 | N=16,062 | 6/3/10,738	0.0008381	2	A05 | N=15,447 | 9/2/11,485	0.0009578
1	220608_MIP-086-CQ_TRT50_TA336	D08 | N=11,223 | 16/3/10,195	0.001864	2	C01 | N=14,184 | 29/0/14,126	0.002053
1	220628_MIP89_trt61_363_20220628_174507_589	B06 | N=19,264 | 151/22/17,163	0.01008	2	A09 | N=16,298 | 13/2/13,567	0.001106
1	220630_MIP088AN_AN02_TA342-7_20220630_131006_932	A05 | N=19,877 | 144/190/7,795	0.04285	2	B01 | N=18,443 | 12/21/6,621	0.004984
1	220704_MIP-90_TRT63_TA367_20220704_150258_016	B09 | N=15,914 | 24/2/13,491	0.001927	2	A01 | N=19,340 | 3/3/9,462	0.0006341
1	220905_MIP-094_TRT20_TA381_20220905_162514_028	D01 | N=17,413 | 0/7/1,992	0.003514	2	H12 | N=18,356 | 3/1/3,667	0.001091

Independent MUT/WT occupancy reliability

Mutant rank	Plate	Known mutant: E(Q1)/E(Q2); Fisher p; warning	Comparison rank	Comparator: E(Q1)/E(Q2); Fisher p; warning
1	211116_TRT63_TA250	33.3/0.7; p=1; OK	2	0.0/0.0; p=1; OK
1	220104_MIP-064_TRT63_TA253	116.7/6.3; p=0.41; OK	2	48.2/7.8; p=0.57; OK
1	220113_TRT76_TA241	12.9/5.1; p=0.43; OK	2	3.8/5.2; p=0.00039; WARN
1	220119_MIP-068-PS_TRT44_TA280	4.0/7.0; p=0.066; OK	2	0.3/0.7; p=1; OK
1	220121_MIP-062-MC_TRT76_TA246	69.4/3.6; p=0.78; OK	2	11.4/0.6; p=1; OK
1	220125_TRT20_TA287	28.4/7.6; p=0.84; OK	2	0.5/2.5; p=1; OK
1	220215_MIP069_TRT19_TA284	1.5/9.5; p=0.65; OK	2	0.6/4.4; p=0.00087; WARN
1	220303_MIP-076-PS_TRT43_TA304	6.4/9.6; p=0.8; OK	2	5.7/1.3; p=1; OK
1	220309_MIP059_TRT42_TA231s_	1.6/8.4; p=0.06; OK	2	1.5/2.5; p=1; OK
1	220324_TRT50_TA299	23.7/7.3; p=0.29; OK	2	16.8/1.2; p=1; OK
1	220331_MIP-072_CQ_TRT23_TA293	6.6/9.4; p=0.31; OK	2	3.7/2.3; p=0.032; OK
1	220408_MIP-067_FF28_TA274	16.7/25.3; p=0.0001; WARN	2	2.2/2.8; p=0.39; OK
1	220427-MIP073CQ_TRT93_TA296_dd	2.4/6.6; p=0.46; OK	2	1.2/3.8; p=0.34; OK
1	220519_MIP_22_088_TRT50__20220518_144839_988	27.0/6.0; p=1.5e-05; WARN	2	9.7/2.3; p=1; OK
1	220520_TRT19_TA330_	1.6/5.4; p=0.2; OK	2	0.0/0.0; p=1; OK
1	220531_MIP-082-MC_TRT76_TA341	61.0/9.0; p=0.37; OK	2	16.2/0.8; p=0.55; OK
1	220531_MIP-085_TRT23_TA333_	6.0/3.0; p=1; OK	2	8.2/2.8; p=0.74; OK
1	220608_MIP-086-CQ_TRT50_TA336	17.3/1.7; p=0.24; OK	2	28.9/0.1; p=1; OK
1	220628_MIP89_trt61_363_20220628_174507_589	155.5/17.5; p=0.25; OK	2	12.5/2.5; p=1; OK
1	220630_MIP088AN_AN02_TA342-7_20220630_131006_932	133.4/200.6; p=0.24; OK	2	11.9/21.1; p=1; OK
1	220704_MIP-90_TRT63_TA367_20220704_150258_016	22.1/3.9; p=0.41; OK	2	2.9/3.1; p=1; OK
1	220905_MIP-094_TRT20_TA381_20220905_162514_028	0.8/6.2; p=1; OK	2	0.8/3.2; p=0.027; OK

N = called droplets; counts are Q1/Q2/Q4. E(Q1/Q2) uses marginal MUT/WT rates. Fisher p < 0.001 is a reliability warning, not an artifact or mutant-call label.

All Labeled Mutants vs. Rank-Based Comparator | Page 3/3

Comparator rule: if mutant rank = 1, show rank 2; otherwise show rank 1.

Observed quadrant counts and MUT/Q4 enrichment

Mutant rank	Plate	Known mutant (well | N | Q1/Q2/Q4)	MUT/Q4	Comparison rank	Comparison well (well | N | Q1/Q2/Q4)	MUT/Q4
1	220929_MIP100_395_3963_20220929_174427_697	H09 | N=19,577 | 34/246/2,873	0.09746	2	G12 | N=18,784 | 2/1/4,050	0.0007407
1	221004_MIP-095_TRT47_TA386_20221005_085757_238	F11 | N=14,868 | 29/51/5,601	0.01428	2	B02 | N=17,088 | 3/4/3,218	0.002175
1	230112_MIP113AO_TA421-7_pI03_20230112_132202_828	H03 | N=20,491 | 14/68/3,802	0.02157	2	F12 | N=20,199 | 26/14/9,081	0.004405
1	230210_MIP093_TRT53_TA378_20230210_063852_538	D04 | N=18,095 | 123/29/14,376	0.01057	2	H06 | N=19,064 | 8/2/13,420	0.0007452
1	230308_MIP-091_TRT27_TA472_20230308_152851_399	F11 | N=18,191 | 6/9/10,612	0.001413	2	G01 | N=19,013 | 3/2/13,555	0.0003689
1	Rapeseed TRT13_TRT005 Aug2019	E10 | N=16,956 | 74/6/16,078	0.004976	2	G08 | N=17,215 | 12/5/14,840	0.001146
2	210212_TRT25-26pooled_TA170	B02 | N=8,273 | 43/0/7,783	0.005525	1	D09 | N=8,908 | 7/9/8,835	0.001811
2	210216_TRT25+26pooled_TRT168	B09 | N=5,960 | 12/0/5,943	0.002019	1	D06 | N=10,970 | 13/1/10,873	0.001288
2	210308_Rescreen of candidates_TA178_TA180_TA175	G01 | N=9,974 | 15/7/7,101	0.003098	1	F12 | N=10,638 | 67/0/10,566	0.006341
2	210316_TRT2+3_TA183	A10 | N=11,997 | 3/26/3,782	0.007668	1	F02 | N=10,408 | 1/27/2,398	0.01168
2	210521_TRT37_TRT105	E07 | N=11,109 | 10/7/6,759	0.002515	1	F07 | N=9,067 | 13/9/6,474	0.003398
2	210525_TRT25+26_TA212	B01 | N=16,693 | 27/0/15,632	0.001727	1	H11 | N=14,891 | 42/0/14,814	0.002835
2	210809_MIP-051_TRT63_TA208c	G01 | N=18,087 | 32/4/16,844	0.002137	1	E03 | N=14,185 | 28/7/12,550	0.002789
2	210810_TRT60_TRT141+1	C06 | N=16,496 | 7/3/10,675	0.0009368	1	A05 | N=15,551 | 3/4/9,589	0.00073
2	210826_TRT5+7_TA218	A08 | N=16,202 | 21/8/10,718	0.002706	1	A10 | N=16,386 | 2/23/1,226	0.02039
2	220621_MIP087-14p_TA357-359_20220621_143802_982	A01 | N=19,173 | 13/11/7,635	0.003143	1	H12 | N=20,300 | 25/15/11,613	0.003444
2	220728_MIP-083-VF_TRT47_TA325_20220728_081450_255	H03 | N=18,428 | 26/19/11,686	0.003851	1	G12 | N=18,367 | 27/28/8,473	0.006491
2	MIP067_seq3-10_314_323	E03 | N=15,242 | 1/53/3	18	1	E02 | N=1,691 | 36/75/279	0.3978
3	200212_TRT13_TRT074	A01 | N=15,539 | 23/7/9,753	0.003076	1	G08 | N=14,683 | 11/14/11,809	0.002117
3	210629_MIP048_TRT51_TA193	E07 | N=17,243 | 28/0/17,207	0.001627	1	A01 | N=16,380 | 54/0/16,305	0.003312
4	210802_MIP-043_TRT63_TA171	C08 | N=14,871 | 7/3/13,236	0.0007555	1	C02 | N=14,435 | 25/0/14,369	0.00174
5	220512_MIP-078-OS_TRT54_TA310	E12 | N=14,532 | 7/2/8,367	0.001076	1	A03 | N=14,501 | 21/0/12,604	0.001666

Independent MUT/WT occupancy reliability

Mutant rank	Plate	Known mutant: E(Q1)/E(Q2); Fisher p; warning	Comparison rank	Comparator: E(Q1)/E(Q2); Fisher p; warning
1	220929_MIP100_395_3963_20220929_174427_697	41.6/238.4; p=0.24; OK	2	0.6/2.4; p=0.12; OK
1	221004_MIP-095_TRT47_TA386_20221005_085757_238	30.3/49.7; p=0.82; OK	2	1.3/5.7; p=0.13; OK
1	230112_MIP113AO_TA421-7_pI03_20230112_132202_828	15.3/66.7; p=0.89; OK	2	18.0/22.0; p=0.016; OK
1	230210_MIP093_TRT53_TA378_20230210_063852_538	121.8/30.2; p=0.92; OK	2	7.0/3.0; p=0.73; OK
1	230308_MIP-091_TRT27_TA472_20230308_152851_399	8.8/6.2; p=0.19; OK	2	3.6/1.4; p=0.63; OK
1	Rapeseed TRT13_TRT005 Aug2019	76.2/3.8; p=0.28; OK	2	14.7/2.3; p=0.073; OK
2	210212_TRT25-26pooled_TA170	40.7/2.3; p=0.17; OK	1	15.9/0.1; p=4.2e-16; WARN
2	210216_TRT25+26pooled_TRT168	12.0/0.0; p=1; OK	1	13.9/0.1; p=0.1; OK
2	210308_Rescreen of candidates_TA178_TA180_TA175	15.7/6.3; p=0.81; OK	1	67.0/0.0; p=1; OK
2	210316_TRT2+3_TA183	9.1/19.9; p=0.015; OK	1	6.5/21.5; p=0.011; OK
2	210521_TRT37_TRT105	10.4/6.6; p=1; OK	1	15.7/6.3; p=0.24; OK
2	210525_TRT25+26_TA212	25.3/1.7; p=0.41; OK	1	41.9/0.1; p=1; OK
2	210809_MIP-051_TRT63_TA208c	33.6/2.4; p=0.3; OK	1	31.0/4.0; p=0.11; OK
2	210810_TRT60_TRT141+1	6.5/3.5; p=1; OK	1	4.3/2.7; p=0.44; OK
2	210826_TRT5+7_TA218	19.2/9.8; p=0.56; OK	1	1.9/23.1; p=0.71; OK
2	220621_MIP087-14p_TA357-359_20220621_143802_982	9.6/14.4; p=0.21; OK	1	22.9/17.1; p=0.53; OK
2	220728_MIP-083-VF_TRT47_TA325_20220728_081450_255	28.6/16.4; p=0.44; OK	1	25.5/29.5; p=0.69; OK
2	MIP067_seq3-10_314_323	0.0/54.0; p=0.014; OK	1	20.7/90.3; p=0.00034; WARN
3	200212_TRT13_TRT074	18.9/11.1; p=0.13; OK	1	20.1/4.9; p=5.6e-05; WARN
3	210629_MIP048_TRT51_TA193	28.0/0.0; p=1; OK	1	53.9/0.1; p=1; OK
4	210802_MIP-043_TRT63_TA171	8.9/1.1; p=0.087; OK	1	24.9/0.1; p=1; OK
5	220512_MIP-078-OS_TRT54_TA310	5.2/3.8; p=0.32; OK	1	18.3/2.7; p=0.1; OK

N = called droplets; counts are Q1/Q2/Q4. E(Q1/Q2) uses marginal MUT/WT rates. Fisher p < 0.001 is a reliability warning, not an artifact or mutant-call label.